

# Grade 8 (Form 2) ICT – Exam Revision & Predicted Questions

Based on the 9 units of the MIE Grade 8 ICT workbook: Computer Operation & Fundamentals · Word Processing · Presentation · Spreadsheet · Internet · Multimedia · Health, Safety & Ethics · Practical Problem Solving & Programming · Database.

*Note: the book itself could not be opened when this was prepared, so notes follow the standard Grade 8 syllabus for each unit. Check each point against your own copy of the book and your teacher's notes.*

## Unit 1: Computer Operation and Fundamentals

- **Data vs information:** data = raw facts/figures; information = processed data that has meaning.
- **Computer system** = hardware + software + user (+ data). Cycle: Input → Processing → Output → Storage.
- **Generations/evolution:** from large machines (vacuum tubes) to transistors, integrated circuits and microprocessors – computers became smaller, faster, cheaper.
- **Binary system:** computers use 0 and 1 (bits). 8 bits = 1 byte. Storage units: KB, MB, GB, TB (each 1024 times the previous).
- **Inside the system unit:** CPU (ALU, control unit, registers), RAM, ROM, motherboard, hard disk/SSD, power supply, ports. CPU speed measured in GHz.
- **Input/Output/Storage devices:** e.g. barcode reader, touchscreen, sensor, plotter, 3D printer; storage: HDD, SSD, USB, optical disc, cloud.
- **Software types:** system (operating system, utilities such as antivirus, backup, disk cleanup) and application (general-purpose, specific/custom). Open source vs proprietary.
- **Operating system:** manages hardware, memory, files, security, multitasking; GUI vs command line.
- **Networks:** LAN, WAN, topology basics (star, bus, ring); devices: router, switch, NIC. Advantages: share resources, files and Internet; disadvantage: security risks, cost.
- **File management:** folders, extensions, copy/move/rename/delete, compress (zip) files, backup.

## Unit 2: Word Processing

- **Revision:** font, alignment, spacing, bullets, tables, headers/footers, page numbers.
- **Page layout:** margins, columns, page breaks, section breaks, watermark, page border.
- **Styles:** heading styles give consistent formatting and allow a **table of contents**.
- **Insert:** images with text wrapping, shapes, SmartArt, WordArt, text boxes, drop cap, hyperlinks, footnotes, symbols.
- **Tables:** insert, merge/split cells, borders, shading, sort data, formulas.
- **Mail merge:** combines one main document with a data source (list of names/addresses) to produce personalised letters, e.g. invitations. Parts: main document, data source, merge fields.
- **Review tools:** spelling & grammar, thesaurus, word count, track changes, comments.
- **Print:** print preview, page range, copies, orientation; save as PDF.
- **Document types:** letter, report, newsletter, flyer, CV.

## Unit 3: Presentation

- **Planning:** purpose, audience, storyboard; one idea per slide; keep text short.
- **Slide master** sets consistent fonts, logo and background for all slides.
- **Objects:** text, images, shapes, charts, tables, audio, video, SmartArt, action buttons, hyperlinks.
- **Animation** (objects) vs **transition** (between slides); timing and order; use sparingly.
- **Navigation:** hyperlinks and action buttons for non-linear presentations; slide sorter view.
- **Design rules:** contrast between text and background, readable font size, limited colours, relevant images, check spelling.
- **Delivery:** rehearse timings, speaker notes, handouts, set up the projector, eye contact.
- **Saving:** .pptx, .ppsx (slide show), PDF.

## Unit 4: Spreadsheet

- **Terms:** workbook, worksheet, cell, range, cell reference, formula bar, fill handle.
- **Cell references:** **relative** (A1, changes when copied), **absolute** (\$A\$1, stays fixed), **mixed** (\$A1 / A\$1). F4 toggles.
- **Functions:** SUM, AVERAGE, MAX, MIN, COUNT, ROUND, IF, COUNTIF, SUMIF. **IF example:** =IF(B2>=50,"Pass","Fail").
- **Order of operations (BODMAS):** brackets first, then \* and /, then + and –.
- **Formatting:** number, currency, percentage, date formats; merge & centre; conditional formatting; borders.
- **Charts:** column/bar, line, pie – choose by purpose; add title, axis labels, legend.
- **Data tools:** sort (A–Z, smallest–largest), filter, freeze panes, find & replace.
- **Common errors:** #DIV/0! (division by zero), #NAME? (wrong function name), #VALUE! (wrong data type), #REF! (invalid reference), ##### (column too narrow).
- **Advantages:** automatic recalculation, accuracy, 'what-if' analysis, quick charts.

## Unit 5: Internet

- **Internet vs WWW; URL** structure: protocol (https) + domain name (www.example.com) + path. **IP address** identifies a device; **DNS** converts domain names to IP addresses.
- **Services:** web, email, chat/VoIP, video conferencing, file transfer, e-commerce, e-banking, e-government, e-learning, cloud computing.
- **Connection types:** dial-up, broadband/ADSL, fibre, Wi-Fi, mobile data (4G/5G). Need: device, ISP, modem/router.
- **Searching:** keywords, quotation marks, Boolean words (AND, OR, NOT), filters; **evaluate sources:** author, date, purpose, bias, accuracy.
- **Email:** To, Cc, Bcc; attachments; folders; signature; netiquette; spam and phishing.
- **Social media:** advantages (communication, sharing) and risks (privacy, cyberbullying, addiction, fake news).
- **Security:** firewall, antivirus, strong passwords, updates, HTTPS padlock, avoid public Wi-Fi for sensitive tasks.
- **Cloud:** store and share files online (Google Drive, OneDrive); collaboration in real time.

- **Advantages/disadvantages of the Internet:** information access and speed vs misinformation, security threats, addiction.

## Unit 6: Multimedia

- **Multimedia** = text + images + audio + video + animation, often **interactive**.
- **Graphics:** bitmap (pixels; JPG, PNG, GIF, BMP) vs vector (shapes; SVG). **Resolution** in dpi/ppi. Colour modes: RGB (screens), CMYK (printing).
- **Audio:** sampling, formats MP3, WAV; trimming, fade in/out, volume; **video:** frames per second, MP4, AVI; editing: cut, join, titles, transitions, sound track.
- **Compression:** **lossy** (smaller file, some quality lost, e.g. JPG, MP3) vs **lossless** (no quality lost, e.g. PNG, ZIP).
- **Animation:** frames shown quickly create movement.
- **Production steps:** plan/storyboard → collect media → edit → assemble → test → publish.
- **Applications:** education, entertainment, advertising, virtual/augmented reality, websites, games.
- **Copyright:** use royalty-free or licensed material; acknowledge sources.

## Unit 7: Health, Safety and Ethics

- **Ergonomics:** posture, screen distance, lighting, breaks; health issues: eye strain, RSI, back pain, stress, obesity from inactivity, addiction.
- **Safety:** electrical hazards (overloaded sockets, liquids, cables), fire safety, safe disposal of old devices (**e-waste** contains toxic materials – recycle).
- **Green computing:** switch off devices, power-saving mode, recycle, print less.
- **Cyber threats:** virus, worm, trojan, spyware, ransomware, hacking, phishing, identity theft, online scams. Protection: antivirus, firewall, updates, backups, strong passwords, caution with links.
- **Ethics:** copyright, plagiarism, software piracy, licences (freeware, shareware, open source), netiquette, respect privacy.
- **Cyberbullying:** do not reply, keep evidence, block, report.
- **Laws in Mauritius:** Data Protection Act, Computer Misuse and Cybercrime Act, Copyright Act.
- **Digital footprint:** traces you leave online – think before posting.

## Unit 8: Practical Problem Solving and Programming

- **Problem solving:** understand → plan/design algorithm → code → test → debug → document/evaluate.
- **Algorithm** shown by **pseudocode** or **flowchart**. Symbols: oval (start/end), rectangle (process), parallelogram (input/output), diamond (decision), arrows (flow).
- **Control structures:** **sequence**, **selection** (if/else), **iteration** (repeat, repeat until, forever, for loop).
- **Variables:** named memory location holding a value; data types: integer, decimal, text (string), Boolean. **Operators:** arithmetic, comparison (>, <, =), logical (AND, OR, NOT).
- **Scratch:** sprites, stage, costumes, blocks (events, motion, looks, sound, control, sensing, operators, variables); create games with score variable and conditions.
- **Errors:** syntax, logic, runtime. **Testing** with sample data; **trace tables** follow variable values step by step.

- **Example algorithm – average of 3 numbers:** Input a, b, c → sum = a + b + c → average = sum / 3 → Output average.

## Unit 9: Database

- **Database** = organised collection of related data stored electronically (e.g. school student records).  
**DBMS** e.g. MS Access, LibreOffice Base.
- **Terms:** **table**, **record** (row = all data about one item), **field** (column = one type of data), **primary key** (unique identifier e.g. Student ID), **data type** (text, number, date/time, currency, yes/no).
- **Objects:** **tables** store data; **queries** search/filter data; **forms** enter data easily; **reports** present/print results.
- **Queries:** use criteria, e.g. Age > 12, Town = "Curepipe"; sort ascending/descending.
- **Advantages over paper files:** fast searching and sorting, less space, avoids duplication, easy updating, secure, accurate reports.
- **Data validation** checks data entered is reasonable (range, type, length, presence). **Data integrity** = accuracy and consistency.
- **Uses:** libraries, hospitals, banks, schools, shops, airlines.
- **Spreadsheet vs database:** databases handle large related data and queries better.

## Key Definitions – Quick Glossary

| Term               | Meaning                                                                      |
|--------------------|------------------------------------------------------------------------------|
| Data               | Raw facts and figures                                                        |
| Information        | Processed data that has meaning                                              |
| Bit / Byte         | Binary digit (0/1) / 8 bits                                                  |
| LAN / WAN          | Network in a small area / over a large area                                  |
| DNS                | Converts domain names to IP addresses                                        |
| Mail merge         | Joining a main document with a data source to produce personalised documents |
| Absolute reference | Cell reference that does not change when copied, e.g. \$A\$1                 |
| Lossy compression  | Smaller file with some quality lost                                          |
| Primary key        | Field that uniquely identifies each record                                   |
| Record / Field     | Row of data about one item / one column of data                              |
| Query              | Request to find data meeting criteria                                        |
| Variable           | Named storage for a value                                                    |
| Trace table        | Table following variable values through an algorithm                         |
| E-waste            | Discarded electronic devices                                                 |

# Predicted Questions

## Section A – Multiple Choice (1 mark each)

1. Raw facts and figures are called:  
A Information B Data C Knowledge D Output
2. How many bits are in a byte?  
A 4 B 8 C 16 D 1024
3. Which device connects computers in a LAN and sends data to the right device?  
A Monitor B Switch C Scanner D Speaker
4. Mail merge needs a main document and a:  
A Table of contents B Data source C Header D Footnote
5. Which reference is absolute?  
A A1 B A\$1 C \$A\$1 D 1A
6. Which formula is correct for a pass mark of 50?  
A =IF(B2>=50,"Pass","Fail") B IF(B2>=50) C =B2>=50 Pass D =PASS(B2)
7. The error #DIV/0! means:  
A Wrong name B Division by zero C Invalid reference D Column too narrow
8. DNS converts:  
A Files to folders B Domain names to IP addresses C Text to audio D Data to records
9. Which compression keeps full quality?  
A Lossy B Lossless C JPG D MP3
10. RGB colour mode is used for:  
A Printing B Screens C Scanning only D Photocopying
11. A software that harms a computer is called:  
A Freeware B Malware C Firmware D Shareware
12. In a database a row is called a:  
A Field B Key C Record D Query
13. A field that uniquely identifies a record is the:  
A Primary key B Report C Form D Filter
14. Which database object is used to search for specific data?  
A Table B Query C Form D Report
15. The flowchart symbol for decision is a:  
A Oval B Rectangle C Diamond D Parallelogram
16. Which control structure repeats instructions?  
A Sequence B Selection C Iteration D Variable
17. Which law protects personal data in Mauritius?  
A Data Protection Act B Road Traffic Act C Fisheries Act D Tax Act

## Section B – Short Answer

1. Differentiate between data and information with an example. [3]

2. Convert the binary number 1011 to decimal (answer 11) and state how many bytes are in 1 KB (1024). [2]
3. Name three components inside the system unit and state the function of the CPU. [4]
4. Differentiate between system software and application software with two examples each. [4]
5. State three advantages and two disadvantages of a computer network. [5]
6. Explain mail merge and give one use. [3]
7. What is the use of heading styles in a long document? [2]
8. State four rules for a good presentation. [4]
9. Differentiate between relative and absolute cell references. [3]
10. Write formulas to: (a) count the numbers in A1:A10, (b) find the largest value in B1:B10, (c) show 'Yes' if C1 is above 100 otherwise 'No'. [3]
11. Give three spreadsheet error messages and their meaning. [3]
12. State what DNS and ISP stand for and their roles. [4]
13. List four ways to evaluate whether a website is reliable. [4]
14. Differentiate between lossy and lossless compression with an example of each. [4]
15. Differentiate between bitmap and vector graphics. [3]
16. What is e-waste? Suggest three ways to reduce its impact. [4]
17. Explain phishing, ransomware and a computer virus. [6]
18. Differentiate between a record, a field and a primary key. [3]
19. State four advantages of a database over paper filing. [4]
20. Name the four main database objects and the function of each. [4]
21. What is data validation? Give two types of validation check. [3]
22. Define algorithm, variable and iteration. [3]

## Section C – Practical / Structured Questions

1. **Word processing:** You must produce a 3-page report on 'Recycling in Mauritius'. Describe how to (a) apply heading styles, (b) add a table of contents, (c) insert a picture with text wrapping, (d) add page numbers and a header, (e) save as PDF. [8]
2. **Spreadsheet:** A teacher stores Name, Maths, Science and English marks for 6 pupils in A1:D7. (a) Write the formula for the total and average of the first pupil. (b) Write an IF formula that shows 'Pass' if the average is 50 or more. (c) Explain how to copy the formula to the other pupils and when to use \$. (d) Which chart compares pupils' totals and why? [10]
3. **Database:** A library stores books. (a) Suggest four fields and their data types. (b) Choose a primary key and explain why. (c) Describe a query to find all books by 'Author = Dickens' published after 2000. (d) State what a report would be used for. [10]
4. **Flowchart:** Draw a flowchart that reads three numbers and outputs the largest. [6]

**5. Trace table:** Trace:  $N = 1$ , total = 0; repeat while  $N \leq 4$ : total = total + N;  $N = N + 1$ ; output total.  
(Answer: total = 10.) [4]

**6. Scratch:** Describe how you would make a catch-the-apple game with a score variable (events, motion, control, variables). [6]

**7. Presentation:** Plan a 6-slide presentation about 'Staying Safe Online' using slide master, a hyperlink/action button and one transition. [6]

**8. Scenario:** You receive an email saying you won a prize and must enter your bank details. (a) Identify the threat. (b) State three actions you should take. (c) Name a law that protects victims in Mauritius. [6]

# Answer Guide

## Section A

1-B, 2-B, 3-B, 4-B, 5-C, 6-A, 7-B, 8-B, 9-B, 10-B, 11-B, 12-C, 13-A, 14-B, 15-C, 16-C, 17-A

## Section B – Model answers

- 1. Data: raw facts (e.g. 45, 60, 72). Information: processed data with meaning (e.g. class average is 59).
- 2.  $1011 = 8+0+2+1 = 11$ ; 1 KB = 1024 bytes.
- 3. CPU, RAM, motherboard (also ROM, hard disk). CPU processes instructions and controls the computer.
- 4. System: Windows, Linux, antivirus. Application: Word, Excel, browser.
- 5. Advantages: share files, printers and Internet; central backup; communication. Disadvantages: security risks, cost, a fault can affect many users.
- 6. Combines a main letter with a list of names/addresses to make personalised copies, e.g. invitations.
- 7. Consistent formatting and allows an automatic table of contents.
- 8. Few words; large readable font; contrasting colours; relevant images; consistent design; limited animations.
- 9. Relative changes when copied (A1). Absolute stays fixed (\$A\$1).
- 10. (a) =COUNT(A1:A10) (b) =MAX(B1:B10) (c) =IF(C1>100,"Yes","No").
- 11. #DIV/0! division by zero; #NAME? unknown function name; #REF! invalid cell reference; #VALUE! wrong data type.
- 12. DNS = Domain Name System, converts domain names to IP addresses. ISP = Internet Service Provider, gives Internet access.
- 13. Check author, date, purpose, source/domain, accuracy compared with other sites, bias.
- 14. Lossy: smaller file but some quality lost (JPG/MP3). Lossless: no quality lost (PNG/ZIP).
- 15. Bitmap: pixels, loses quality when enlarged. Vector: mathematical shapes, scales without loss.
- 16. Discarded electronic devices. Recycle, repair/reuse, donate, buy durable devices.
- 17. Phishing: fake messages stealing data. Ransomware: locks files and demands payment. Virus: program that spreads and damages files.
- 18. Record: row for one item. Field: column of one data type. Primary key: unique field such as ID.
- 19. Fast search, less space, less duplication, easy updating, accurate reports, better security.
- 20. Tables store; queries search; forms enter data; reports print/present.
- 21. Checking entered data is sensible; e.g. range check, type check, length check, presence check.
- 22. Algorithm: ordered steps to solve a problem. Variable: named storage for a value. Iteration: repeating steps (loop).

## Section C – Hints

- Q2: Total =SUM(B2:D2); Average =AVERAGE(B2:D2); =IF(E2>=50,"Pass","Fail"); drag the fill handle; use \$ to fix a cell such as a pass mark; column chart compares individuals.

- Q3: Fields: BookID (text/number), Title (text), Author (text), Year (number). Primary key BookID since unique. Query criteria: Author = "Dickens" AND Year > 2000.
- Q4: Input A,B,C → if A>B and A>C output A, else if B>C output B, else output C.
- Q5: Iterations: N=1 total 1; N=2 total 3; N=3 total 6; N=4 total 10; N=5 stop. Output 10.
- Q8: Phishing; do not click or reply, delete/report, change passwords if needed; Computer Misuse and Cybercrime Act.

**Exam tips:** Read each question twice · Use marks to decide how many points to give · Practise formulas and flowcharts · Revise definitions · Show your working · Manage your time.